

Scientific Skills Exercise

Interpreting a Scatter Plot with a Regression Line

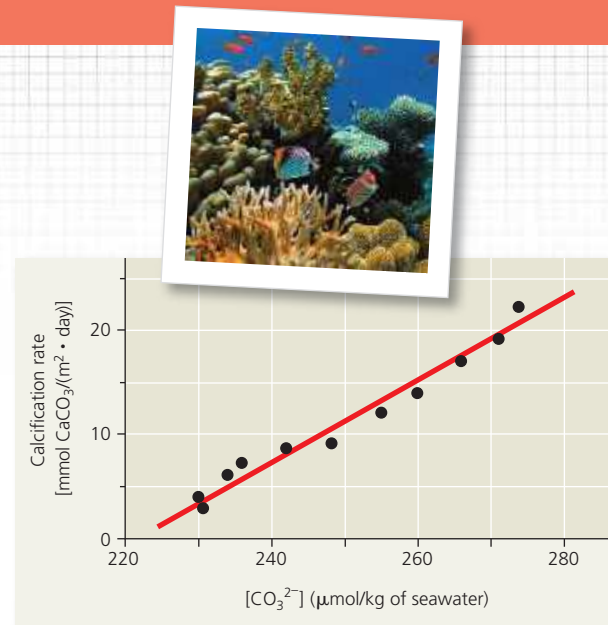
How Does the Carbonate Ion Concentration of Seawater Affect the Calcification Rate of a Coral Reef? Scientists predict that acidification of the ocean due to higher levels of atmospheric CO_2 will lower the concentration of dissolved carbonate ions, which living corals use to build calcium carbonate reef structures. In this exercise, you will analyze data from a controlled experiment that examined the effect of carbonate ion concentration ($[\text{CO}_3^{2-}]$) on calcium carbonate deposition, a process called calcification.

How the Experiment Was Done For several years, scientists conducted research on ocean acidification using a large coral reef aquarium at Biosphere 2 in Arizona. They measured the rate of calcification by the reef organisms and examined how the calcification rate changed with differing amounts of dissolved carbonate ions in the seawater.

Data from the Experiment The black data points in the graph form a scatter plot. The red line, known as a linear regression line, is the best-fitting straight line for these points.

INTERPRET THE DATA

- When presented with a graph of experimental data, the first step in analysis is to determine what each axis represents. (a) In words, what is shown on the x-axis? (Include the units.) (b) What is on the y-axis? (c) Which variable is the independent variable—the one that was *manipulated* by the researchers? (d) Which is the dependent variable—the one that responded to or depended on the treatment, which was *measured* by the researchers? (For additional information about graphs, see the Scientific Skills Review in Appendix D.)
- Based on the data shown in the graph, describe in words the relationship between carbonate ion concentration and calcification rate.
- (a) If the seawater carbonate ion concentration is $270 \mu\text{mol/kg}$, estimate the rate of calcification and how many days it would



Data from C. Langdon et al., Effect of calcium carbonate saturation state on the calcification rate of an experimental coral reef, *Global Biogeochemical Cycles* 14:639–654 (2000).

- take 1 square meter of reef to accumulate 30 mmol of calcium carbonate (CaCO_3). (b) If the seawater carbonate ion concentration is $250 \mu\text{mol/kg}$, what is the approximate rate of calcification, and approximately how many days would it take 1 square meter of reef to accumulate 30 mmol of calcium carbonate? (c) If the carbonate ion concentration decreases, how does the calcification rate change, and how does that affect the time it takes coral to grow?
- (a) Which step of the process in Figure 3.12 is measured in this experiment? (b) Are the results of this experiment consistent with the hypothesis that increased atmospheric $[\text{CO}_2]$ will slow the growth of coral reefs? Why or why not?

➔ **Instructors:** A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

3 Chapter Review



➔ Go to **Mastering Biology** for Assignments, the eText, the Study Area, and Dynamic Study Modules.

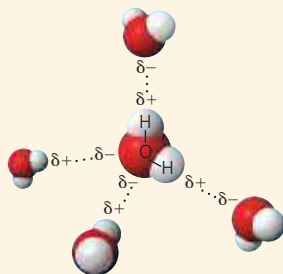
SUMMARY OF KEY CONCEPTS

➔ To review key terms, go to the **Vocabulary Self-Quiz** in the **Mastering Biology** eText or Study Area, or go to goo.gl/zkjj9t.

CONCEPT 3.1

Polar covalent bonds in water molecules result in hydrogen bonding (p. 45)

- Water is a **polar molecule**. A hydrogen bond forms when a partially negatively charged region on the oxygen of one water molecule is attracted to the partially positively charged hydrogen of



a nearby water molecule. Hydrogen bonding between water molecules is the basis for water's properties.

DRAW IT Label a hydrogen bond and a polar covalent bond in the diagram of five water molecules. Is a hydrogen bond a covalent bond? Explain.

CONCEPT 3.2

Four emergent properties of water contribute to Earth's suitability for life (pp. 45–50)

- Hydrogen bonding keeps water molecules close to each other, giving water **cohesion**. Hydrogen bonding is also responsible for water's **surface tension**.
- Water has a high **specific heat**: Heat is absorbed when hydrogen bonds break and is released when hydrogen bonds form. This helps keep **temperatures** relatively steady, within limits that permit life. **Evaporative cooling** is based on water's high **heat**